

Adaptive Physics-Informed Thermal Digital Twins: PINN and Streamed-PINO Models for Rocket-Wall Heat Conduction

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This work develops a causal thermal digital twin for a regeneratively cooled rocket-throat wall using physics-informed neural networks (PINNs) and a streamed physics-informed neural operator (PINO). A one-dimensional through-thickness prototype first verifies the governing physics, reference solver, and sensor-assimilation pipeline. The same diffusion physics is then extended to a two-coordinate analytical benchmark and to causal boundary-change tests. The adaptive PINN continues optimization as sparse measurements arrive; the streamed PINO instead freezes its physics-informed base operator and updates a compact recurrent observer. On a held-out heat-flux change from 4.302 to 5.189 MW/m² at $t = 0.53$ s, overall RMSE is 21.960 K for the offline PINN, 2.180 K for the adaptive PINN, and 0.226 K for the streamed PINO. PINO post-switch RMSE is 0.189 K, while median online inference is on the millisecond scale rather than the adaptive PINN’s 2.26-s optimization update. The results show that sparse causal assimilation substantially improves disturbance tracking and that an offline-trained operator with a recurrent correction can provide the strongest accuracy–latency compromise.

I. Introduction

Thermal protection and regenerative cooling depend on estimating temperatures at locations where instrumentation is sparse or impractical. A digital twin can combine a governing heat-transfer model with wall measurements to reconstruct the complete temperature field. Classical numerical solvers offer reliable predictions when geometry, properties, and boundary conditions are known, whereas purely data-driven surrogates can update quickly but may extrapolate unphysically. PINNs embed the governing differential equation in the training objective and thereby offer a continuous field representation constrained by physics [1, 2].

The central question is not simply whether a PINN can solve a heat equation, but whether a thermal model can in-

corporate a stream of sparse measurements when operation changes. Two causal strategies are compared. The adaptive PINN continues optimization using a rolling observation window. The streamed PINO learns a family of heat-solution operators offline and updates only a recurrent observer state online. Both are evaluated using only measurements available at the current time.

The 1D and 2D studies are a verification hierarchy for the same conduction physics, not unrelated applications. The 1D coordinate resolves wall thickness; the second coordinate resolves in-plane or circumferential variation. The Cartesian square benchmark supplies a separable analytical reference from Hsu, Tu, and Chang [3], while the wall model retains dimensional heat flux, convection, and temperature. The source and reproducibility instructions are available in the project repository [6].

II. Physical Problem Definition

A. One-dimensional rocket-wall prototype

The prototype represents conduction through a wall of thickness L . The hot surface at $x = 0$ receives a prescribed heat flux and the coolant surface at $x = L$ obeys a convective condition:

$$\rho c_p T_t = k T_{xx}, \quad 0 < x < L, \quad 0 < t < t_f, \quad (1)$$

$$-k T_x(0, t) = q_{\text{hot}}(t), \quad (2)$$

$$-k T_x(L, t) = h_{\text{cool}}[T(L, t) - T_{\text{cool}}], \quad (3)$$

$$T(x, 0) = T_0. \quad (4)$$

Constant copper-like properties and no volumetric generation are assumed. A Crank–Nicolson solution and, in the final physical study, a Robin eigenfunction expansion supply synthetic ground truth. The 1D equation is the locally planar, through-thickness reduction of the wall problem. For an axisymmetric cylindrical wall the radial term is $r^{-1} \partial_r (r \partial_r T)$; over a thin wall with $L/R \ll 1$, r varies weakly and curvature enters as a nearly constant geometric correction. The 2D Cartesian benchmark adds a tangential coordinate while retaining the same parabolic diffusion operator.

B. Two-dimensional analytical benchmark

The extended study uses the dimensionless heat equation on a unit square,

$$\theta_\tau = L_r^2 \theta_{XX} + \theta_{YY}, \quad (X, Y) \in (0, 1)^2, \quad (5)$$

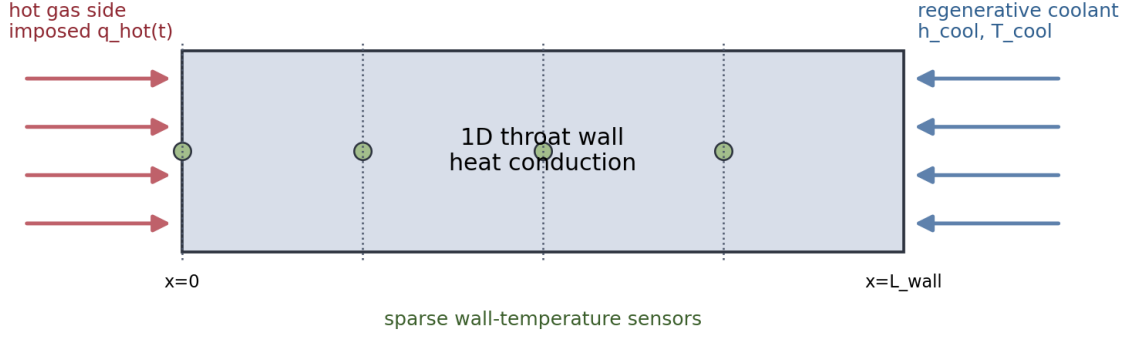


Fig. 1 Unified wall interpretation. The 1D model resolves through-thickness conduction; the extended model additionally resolves variation along the wall. Sensors are observation constraints, not boundaries.

with $L_r = 1$. For literature verification, the four edges have parabolic spatial profiles and independent exponential decay rates:

$$\begin{aligned} \theta(0, Y, \tau) &= (Y - Y^2)e^{-d_1\tau}, & \theta(1, Y, \tau) &= (Y - Y^2)e^{-d_2\tau}, \\ \theta(X, 0, \tau) &= (X - X^2)e^{-d_3\tau}, & \theta(X, 1, \tau) &= (X - X^2)e^{-d_4\tau}, \end{aligned} \quad (6)$$

with $\theta(X, Y, 0) = (X - X^2) + (Y - Y^2)$. The analytical solution is evaluated by shifting the nonhomogeneous boundaries and expanding the homogeneous remainder in sine eigenfunctions [3].

The adaptive experiment instead applies continuous left-edge heating $\theta(0, Y, \tau) = Y(1 - Y)$ and zero temperature on the other edges. At $\tau_s = 50$, the heating amplitude rises by 50%. The event is excluded from the PINN boundary definition, so only subsequently revealed measurements communicate the disturbance. The analytical switch solution preserves the pre-event interior field through a homogeneous sine-series correction while enforcing the new edge values.

III. Methodology

A. PINN formulation and verification ladder

The network maps normalized coordinates to temperature. Automatic differentiation supplies the derivatives in Eq. (5). The offline objective is

$$\mathcal{L}_0 = w_f \mathcal{L}_f + w_b \mathcal{L}_b + w_i \mathcal{L}_i, \quad (7)$$

where the three mean-square terms penalize the PDE residual, boundary conditions, and initial condition. The adaptive objective adds sparse observations,

$$\mathcal{L}_{\text{ad}} = \mathcal{L}_0 + w_d \frac{1}{N_s} \sum_{j=1}^{N_s} [\theta_{\text{PINN}}(\mathbf{x}_j, \tau_j) - \theta_j]^2. \quad (8)$$

Verification is deliberately layered. Unit tests cover the analytical solution, architecture configuration, batching, metrics, and boundary-switch solver. The 1D neural prediction is checked against Crank–Nicolson. The 2D analytical evaluator is checked against the three tables of Ref. [3]. Table 1 is reproduced closely; direct use of the printed equations does not reproduce printed Tables 2 and 3, so only Table 1 is treated as a verified benchmark. Finally, every DeepXDE run stores the resolved TOML configuration beside its metrics and figures.

B. Causal online assimilation

After offline training, sensor values are revealed in chronological batches. Immediately before assimilation, the current model predicts the newly available points and full analytical field. Those predictions define the causal-prior error. The observations in a recent-history window are then attached as point constraints, and Adam continues from the current weights for a fixed iteration budget. Re-evaluation defines the causal-posterior error. This distinction prevents a final network from being evaluated retrospectively and misrepresented as an online forecast.

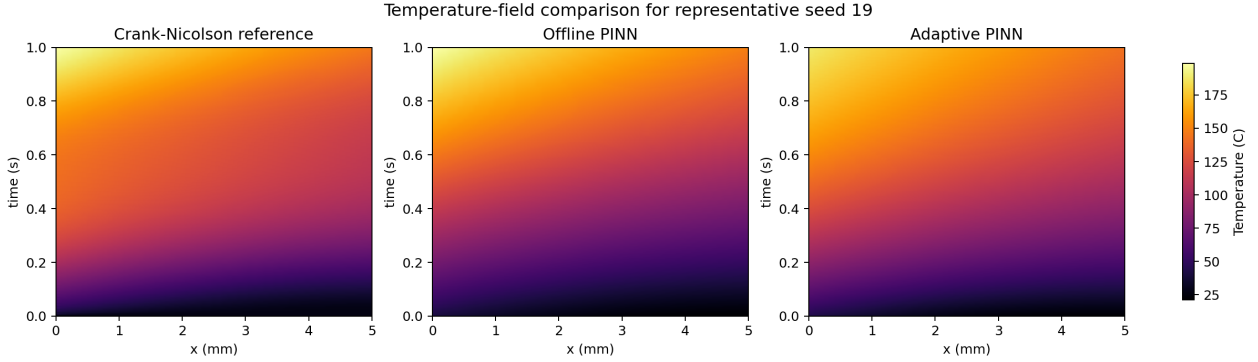
Three controlled configurations use the same PDE and reference field. The balanced and low-latency models contain three 48-neuron tanh layers, 2,500 offline Adam iterations, and 100 iterations per online update. Balanced assimilation uses three sensors, spacing $\Delta\tau = 2.5$, two retained batches, and $w_d = 10$. Low latency uses a 5×5 grid, $\Delta\tau = 1.25$, four retained batches, and $w_d = 20$. The PyTorch-comparison preset matches the standalone network and analogous training choices while retaining the 2D equation. Table 1 summarizes the replaceable choices.

C. Streamed physics-informed neural operator

Unlike an instance-specific PINN, an operator network learns a mapping over a family of boundary histories [5]. The implemented streamed PINO uses a frozen parameter-conditioned physics-informed base operator and a causal correction model. At time t_k , nine sensor values on a 3×3 grid are compared with the base prediction. A gated recurrent unit (GRU) maps this innovation, the measurements, boundary parameters, and time into a 96-value state. A spatial decoder converts the state into a correction that is multiplied by $x(1-x)y(1-y)$, preserving the imposed boundary values. Network weights remain frozen online; only the recurrent state evolves. The shared dataset contains 200 training, 40 validation, and 100 interpolation-test boundary histories on 17×17 fields, plus one locked switch sequence. All fields are generated independently from the 20-term analytical series, and no future or full-field test data enter the online API.

Table 1 Two-dimensional adaptive configurations.

Setting	Balanced	Low latency	PyTorch match
Sensors	3	25	9
$\Delta\tau$	2.5	1.25	2.5
Update windows	21	81	5
Recent batches	2	4	5
Data weight	10	20	10
Runtime [s]	201.5	304.0	42.7

**Fig. 2 One-dimensional Crank–Nicolson reference, offline PINN, and adapted PINN for a representative seed.**

IV. Results and Discussion

A. One-dimensional prototype

The PyTorch prototype establishes that the implementation satisfies the intended heat equation, boundary conditions, nondimensionalization, and streaming-data path before increasing dimensionality. Its Crank–Nicolson reference was separately checked against the constant-flux analytical steady state, agreeing within 0.007 K. The preliminary short-budget runs also exposed a useful training lesson: a 350-epoch PINN retained a smooth cold bias near the flux boundary, whereas longer Adam training reduced the global error monotonically. Consequently, the preliminary under-trained comparison is treated as a development diagnostic rather than a final performance claim. Figure 2 is retained to show the verified field structure and model progression; quantitative conclusions use the reproducible final boundary-change benchmark in Table 2.

B. Nominal two-dimensional PINN operation

The two-dimensional results are less uniformly favorable and therefore more informative. The balanced model obtains offline RMSE 0.06193, while its final adaptive RMSE is 0.06814, a 10.0% degradation. Only 15 of its 21 update windows improve the local pre/post-update comparison. In contrast, low latency reduces global RMSE from 0.05944 to 0.05474 (7.9%) and improves 60 of 81 windows. The PyTorch-matched configuration reduces RMSE from

Table 2 Held-out random boundary-change performance.

Model	Overall	Pre-switch	Post-switch	Online latency
		RMSE [K]		[ms]
Offline PINN	21.960	15.682	27.494	–
Adaptive PINN	2.180	2.966	0.263	2262.8
Streamed PINO	0.226	0.254	0.189	~2

0.08625 to 0.07228 (16.2%) and improves four of five windows.

The balanced failure is a useful negative result. Sparse point fitting can improve observed locations while disturbing a globally adequate physics solution. Rolling history also creates a changing optimization target, and equal PDE/boundary/initial weights do not guarantee that a larger data weight preserves every constraint. Thus final global RMSE alone is insufficient; causal error traces and the number of helpful windows expose when adaptation is actually useful.

C. Unannounced heat-flux change and PINO comparison

The stronger digital-twin test uses a held-out heat flux of 4.302 MW/m^2 followed by an unannounced increase to 5.189 MW/m^2 at $t = 0.53 \text{ s}$. Table 2 reports dimensional full-field errors. The adaptive PINN reduces overall RMSE by 90.1%, from 21.960 to 2.180 K. The streamed PINO reduces it by 99.0%, to 0.226 K, and maintains post-switch RMSE of 0.189 K. A second unseen, no-break case at 4.596 MW/m^2 produces PINO RMSE 0.226 K, demonstrating that the result is not restricted to a single switch. Across 100 interpolation scenarios, PINO’s dimensionless RMSE is 0.000260 ± 0.000093 .

The latency distinction is architectural. The adaptive PINN performs gradient optimization after new measurements and therefore requires a median 2.26 s per update in this test. The PINO performs one frozen-base evaluation and one recurrent-state update, giving approximately 2 ms latency in the report-quality runs. Figure 4 shows that its pointwise error remains below 1 K over almost the entire wall-time domain.

D. Limitations

All observations are synthetic, so the study does not include calibration error, sensor bias, uncertain properties, or geometry mismatch. Cartesian and thin-wall models neglect full nozzle curvature and conjugate coolant flow; the cylindrical reduction is accurate only when curvature variation across the wall is small. Runtime measurements are hardware-dependent. Although PINO is evaluated on 100 interpolation cases and two held-out physical cases, broader extrapolation in heat flux, material properties, noise, and switch timing remains untested. The discrepancy in published Tables 2 and 3 further illustrates why reference data must be independently verified.

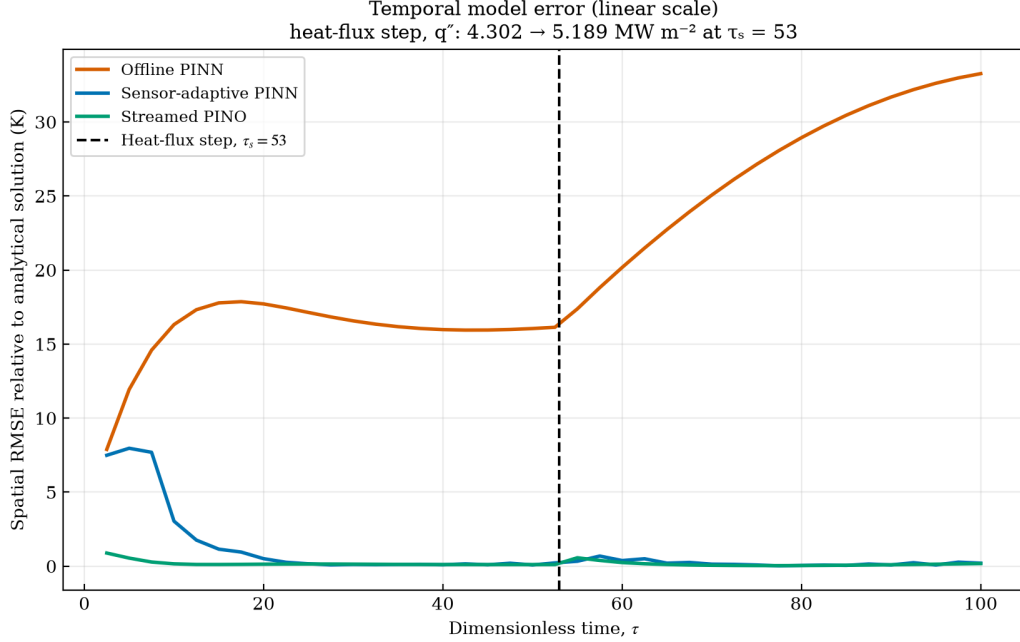


Fig. 3 Readable standalone RMSE comparison for the held-out random boundary change. The vertical marker denotes the heat-flux switch at $t = 0.53$ s.

V. Conclusions

This project demonstrates an end-to-end thermal digital twin through a unified verification hierarchy. The through-thickness prototype verifies wall physics and sparse assimilation; the two-coordinate analytical benchmark verifies the extended diffusion solver and exposes a literature inconsistency. Under a held-out heat-flux change, adaptation reduces PINN error from 21.960 to 2.180 K, while the streamed PINO reaches 0.226 K overall RMSE and millisecond online latency.

The main critical insight is that online adaptation is not automatically beneficial: a sparse balanced PINN configuration degrades nominal global accuracy even though most individual updates help. Operator learning changes this tradeoff by moving expensive learning offline and maintaining only a causal observer state during operation. Future work should introduce experimental thermocouple data, uncertain properties, noise and placement sweeps, and full cylindrical or conjugate heat-transfer geometry.

Author Contributions

Gabriel Berezovsky, Joshua Blecher, and Nathan Florsheim contributed to the physical formulation, software implementation and verification, analysis of figures and metrics, and manuscript preparation. The three-person team was completed with prior course approval.

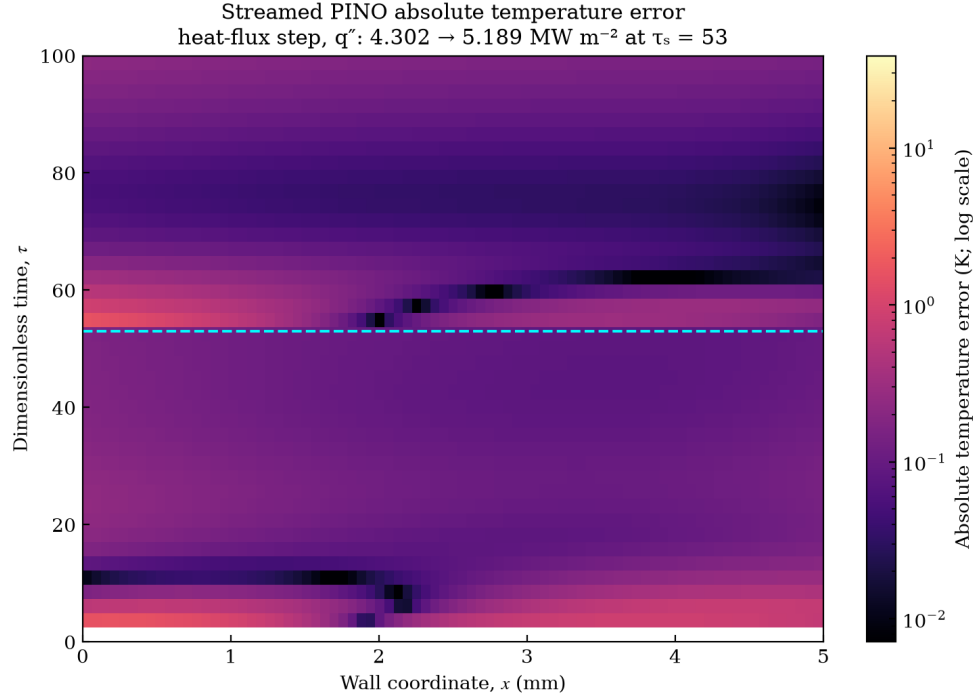


Fig. 4 Standalone streamed-PINO absolute-error field for the held-out random boundary change; the dashed line marks the switch.

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